Master: Amylase Rate, Graph and Topic 1 Consolidation

Today I can… ☐ Show your working: relative rate = 1 ÷ time. Use the unit s^{-1}. ☐ Read or plot how pH links to rate. Say only what the data supports. ☐ Name a limit of the method. Link it to a way to improve it. Then show how digestion differs from diffusion.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Lesson strip

Today I can

Show your working: relative rate = 1 ÷ time. Use the unit s^{-1}.

Read or plot how pH links to rate. Say only what the data supports.

Name a limit of the method. Link it to a way to improve it. Then show how digestion differs from diffusion.

In your book

In your book: Glue in today's strip; date it. Answer the four clues.

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Arrival · Supported

1. Time and rate

With the same starting starch and endpoint, which is faster: 90 s or 180 s?

Shorter time for the same measured change means faster breakdown.

2. Read the endpoint

Link to today: Blue-black means starch detected. Orange-brown means starch not detected. Which colour meets our endpoint?

Endpoint = first sampled time with starch not detected.

3. Show the working

Relative rate = 1 ÷ time. Use a calculator to find 1 ÷ 100.

Press 1 ÷ 100 =. Use time in seconds; rate unit is s^{-1}.

4. Match the instruction

DESCRIBE means state the pattern. EVALUATE means judge the evidence. Which asks for a strength and a limitation?

The command word tells you what the answer must do.

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Exit Ticket · Supported

Evidence

Illustrative matched-condition data: pH 6 endpoint 120 s; pH 7 endpoint 90 s. A separate pH 7 repeat gave 120 s. Relative rate = 1 ÷ time in seconds (s^{-1}). Samples were 30 s apart; pH 4–8 were tested.

1. Calculate the pH 6 relative rate, then choose the faster first-series result: pH 6 or pH 7.

Use 1 ÷ 120. For matched endpoints, shorter time means faster breakdown.

2. Choose the stronger conclusion: “pH 7 was fastest in this first series” or “pH 7 is best for every amylase”. Give one clue.

Consider what was tested and the separate repeat.

Under the heading Exit ticket: answer the short questions in your book.

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Arrival · Independent

1. Time and rate

Explain why 90 s gives a larger relative rate than 180 s under matched conditions.

Relative rate = 1 ÷ time, for the same starting amount and endpoint.

2. Read the endpoint

At 60 s starch is detected; at 90 s it is not. What endpoint value is used in the rate calculation?

Use the first observed endpoint, not an invented time between samples.

3. Show the working

Calculate relative rate for an endpoint of 120 s; show the equation, substitution and unit.

Relative rate = 1 ÷ time. A value to three significant figures is sufficient here.

4. Match the instruction

Distinguish DESCRIBE from EXPLAIN when answering about a graph.

Describe = state the pattern. Explain = link that pattern to scientific reasons.

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Exit Ticket · Independent

Evidence

Illustrative matched-condition data: pH 6 endpoint 120 s; pH 7 endpoint 90 s. A separate pH 7 repeat gave 120 s. Relative rate = 1 ÷ time in seconds (s^{-1}). Samples were 30 s apart; pH 4–8 were tested.

1. Work out both relative rates from the first series. Use them to compare the results. Set a limit on what this shows.

Show 1 ÷ time and s^{-1}. End with “in this series”.

2. Use the repeat or the sampling interval. Give one limit of the test. Then give one change for the next test. Say why it would help.

Name the limitation before an improvement. A shorter interval must be authorised and feasible.

Under the heading Exit ticket: answer the short questions in your book.

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Arrival · Aspire

1. Time and rate

A trial takes twice as long to reach the same endpoint. What happens to its reciprocal relative rate?

Compare 1/t with 1/(2t). Starting amount and endpoint must be matched.

2. Read the endpoint

Why is 1/90 s^{-1} an endpoint-based estimate rather than a direct instantaneous rate measurement?

The endpoint records one sampled stage of change; rate may vary during a reaction.

3. Show the working

For 120 s, compare 1/t with 1000/t. May the two values share one unlabelled graph axis?

The two conventions differ by a scale factor of 1000. This lesson uses 1/t only.

4. Match the instruction

A graph has one result at each pH. Name one extra thing an EVALUATE answer needs beyond describing its peak.

Evaluate = use a strength or limitation to reach a supported judgement.

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Exit Ticket · Aspire

Evidence

Illustrative matched-condition data: pH 6 endpoint 120 s; pH 7 endpoint 90 s. A separate pH 7 repeat gave 120 s. Relative rate = 1 ÷ time in seconds (s^{-1}). Samples were 30 s apart; pH 4–8 were tested.

1. Does the pH 7 repeat strengthen the claim of an exact optimum? Use a rate and a comparison.

Its 120 s result is a separate observation at the same pH. Compare with pH 6.

2. Give one change to the method that would help with a specific problem. Say why this graph still cannot prove permanent enzyme denaturation.

Better endpoint evidence is one thing. Direct tests for a change in the molecules, or for recovery, are another. Show how these differ.

Under the heading Exit ticket: answer the short questions in your book.

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Starter and method strip

Starter

Matched amounts, same iodine endpoint: reaction A reaches the endpoint in 90 s; B takes 180 s. Which is faster?

Method strip

1 · Find the endpoint time that was measured: 120 s.

2 · Put that time in the sum: relative rate = 1 ÷ 120.

3 · Work out the sum: 0.008333… s^{-1}.

4 · Write the result to 3 significant figures: 0.00833 s^{-1}.

Master: Amylase Rate, Graph and Topic 1 Consolidation

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You Do · Independent

Calculate relative rate for pH 6 and pH 7, showing equation, substitution and units. Check that shorter time gives a larger value. [4 marks]

Complete the five-point graph using the two calculated rates and three supplied points. Label both axes and state the rise-and-fall pattern. [4 marks]

Identify the highest tested activity in the first series using one numerical comparison. Explain how pH can affect the enzyme's active site. [3 marks]

Give one strength, one limitation and one linked improvement for the endpoint method. Use the separate pH 7 repeat of 120 s to qualify your confidence. [4 marks]

Distinguish enzyme digestion from diffusion in the supplied two-step diagram. Explain why movement from high to low concentration does not itself break starch into sugars. [3 marks]

I · S · C

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You Do · Supported

1. Use 1 ÷ time to calculate the pH 6 rate from 120 s and the pH 7 rate from 90 s.

Calculator steps: 1 ÷ 120 =; then 1 ÷ 90 =. Keep the unit s^{-1}.

2. Add pH 6 and pH 7 to the supplied rate graph. Match each pH to its calculated rate; the other three points are supplied.

Read pH along the bottom, then move up to the rate. Use the labelled example point.

3. Choose the highest rate in the first series. Complete “In these tested conditions, pH ___ was fastest because ___.”

Find the largest rate number, not the largest pH number. Include “in this series”.

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Feedback: □ VF □ WS □ I □ NS □ E □ R □ [//] □ [?] _____

You Do · Supported · continued

4. Samples are taken 30 s apart. Choose a limit of the method and a linked way to improve it: uncertain exact endpoint / use a shorter fixed interval if authorised.

Say “Because samples are separated by ___ s, we cannot know ___.”

5. Look at the diagram given. First, starch breaks into sugars. Then particles move from high to low concentration. Label the two changes: digestion / diffusion.

Digestion changes molecules in a chemical way. Diffusion changes where the particles are.

I · S · C

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You Do · Choice

Match 120 s and 90 s to 0.00833 s^{-1} and 0.0111 s^{-1}. Show 1 ÷ time for each. Add the pH 6 and pH 7 points to the supplied graph; choose its pattern: rises then falls / always rises. Record C beside your book response.

Choose the highest first-series rate: pH 4 / pH 7 / pH 8. Give one number. Choose how pH may change activity: active-site fit / enzyme choice. Match the 30 s sampling gap to uncertain exact endpoint / exact timing; choose a shorter authorised interval / invented time as the improvement.

The pH 7 repeat took 120 s. Choose confidence: variation needs repeats / one exact optimum is proved. Match starch breakdown / net particle movement to digestion / diffusion. Write one sentence explaining the difference.

After choosing, write one short scientific sentence, label or reason in your book.

I · S · C

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You Do · Aspire

Calculate pH 6 and pH 7 relative rates. Compare the first pH 7 rate with its separate 120 s repeat, retaining both results. [4 marks]

The repeat is a second observation at pH 7. Do not just take a mean of times or rates. If you choose that method, say so and explain it.

Complete all five points on the first-series graph. Add the separate repeat at the same pH 7 x-position. Use a different symbol for it. Explain why it is not pH 9. Explain why it must not just replace a point without saying so. [4 marks]

Same independent variable means the same x-position. Use a key for the repeat.

Evaluate “pH 7 is the exact optimum for every amylase”. Use the tested range, the repeat and the difference between data and a molecular explanation. [4 marks]

The highest point in this first series is a smaller claim than one exact optimum for all amylase. Judge how precise the evidence is. Judge how far the claim can go.

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You Do · Aspire · continued

Give a strength–limitation–improvement chain. Explain which uncertainty a shorter authorised sampling interval reduces and which it leaves. [4 marks]

Target a named source of uncertainty; “repeat it” alone does not explain a smaller sampling gap.

Explain why “diffusion digests starch” mixes up two processes. Use the definitions given. State why an enzyme-rate graph on its own cannot show how sugars cross a gut membrane. [4 marks]

Think about what the graph measured. Use this to say what it cannot show. Starch disappearing does not tell us how a membrane works.

I · S · C

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You Do · shared evidence

Evidence and reference diagrams

Enzyme: biological catalyst; speeds a reaction without being used up. Substrate: substance it acts on. pH: a measure of acidity or alkalinity. Amylase acts on starch, producing smaller sugars.

pH | Endpoint / s | Relative rate / s^{-1}

4 | 240 | 0.00417

5 | 180 | 0.00556

6 | 120 | 0.00833

7 | 90 | 0.0111

8 | 150 | 0.00667

Supplied illustrative observations — not your practical results. Single runs; equal starting amounts and the same iodine endpoint. Samples every 30 s. pH 7: blue-black at 30/60 s, orange-brown at 90 s. Record 90 s.

Complete the reference graph: pH 4, 5 and 8 are shown; add pH 6 and pH 7. Aspire: add the separate repeat with a distinct symbol.

Rate: 1 ÷ endpoint time. At pH 7: 1 ÷ 90 = 0.0111 s^{-1} (3 significant figures). These are single-run rates, not means. Highest among tested: pH 7; exact optimum is not established. The extra pH 7 repeat (120 s) is a separate trial, not silently averaged.

Topic 1 transfer: Digestion breaks large molecules into smaller ones. Diffusion is net particle movement. It goes from a higher to a lower concentration. It does not break up a starch molecule. Enzymes are catalysts. They are not living things. A change in pH can change how well they work. A slower time alone does not prove permanent denaturation.

Describe = give a pattern; explain = link cause and effect; evaluate = use evidence and limits. An unobserved endpoint by 300 s establishes no exact reciprocal rate.

Master: Amylase Rate, Graph and Topic 1 Consolidation

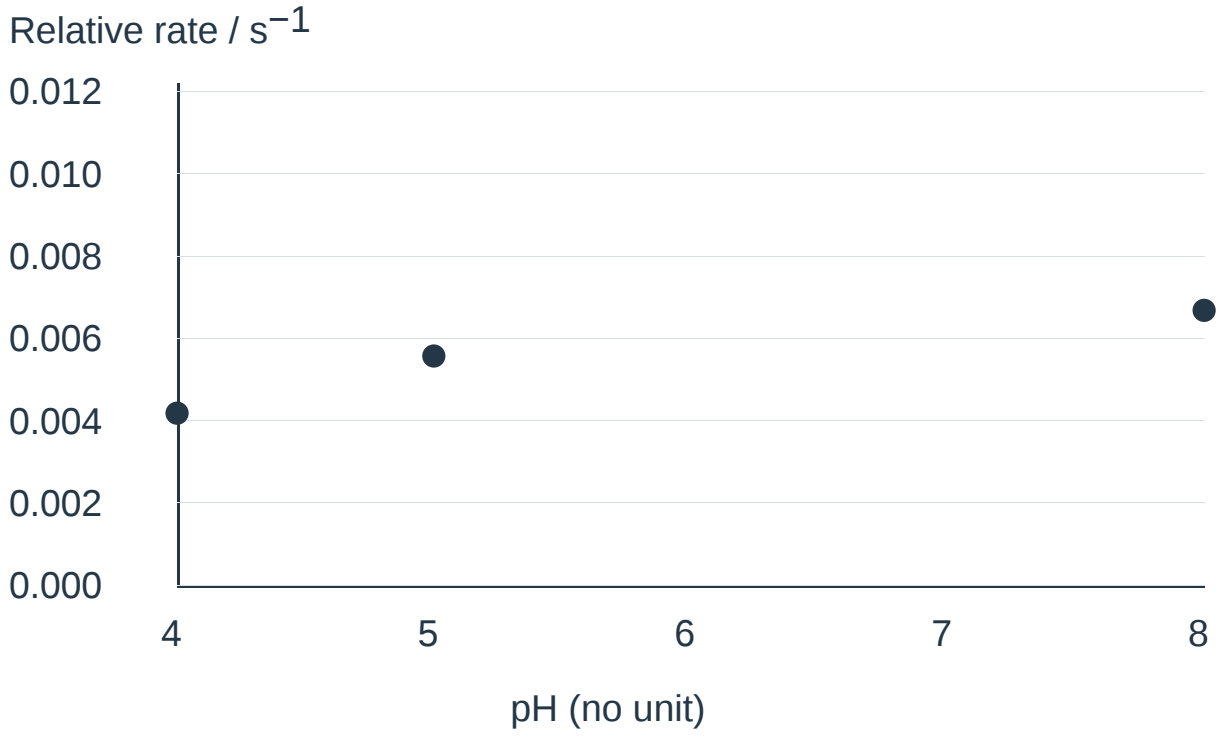
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You Do · shared evidence · continued

Reference

Enzyme: biological catalyst; speeds a reaction without being used up. Substrate: substance it acts on. pH: a measure of acidity or alkalinity. Amylase acts on starch, producing smaller sugars.



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You Do · shared evidence · continued

Digestion and diffusion

Enzyme digestion changes molecules in a chemical way. It breaks large molecules into smaller ones. Diffusion is net movement of particles. They move from higher to lower concentration. That movement does not itself break starch into sugars in a chemical way.

Amylase speeds up starch digestion. It helps the starch break down. Just moving particles to a new place does not split their molecules. These are two different sorts of change.

Digestion: chemical breakdown

Starch chain → smaller sugars

Diffusion: net movement



Higher → lower concentration

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Knowledge organiser

Learning goal

I can calculate and graph relative amylase rates, judge what the evidence supports, and use enzyme and transport ideas in an exam-style explanation.

Calculate a reciprocal

Relative rate = 1/time. For 120 s: 1/120 = 0.00833 s^{-1}. For 90 s: 1/90 = 0.0111 s^{-1}. A shorter endpoint time gives a larger reciprocal, for matched amounts and the same stopping rule.

Describe then qualify

These illustrative first-series points rise from pH 4 to 7 and fall at pH 8. pH 7 is highest among the values tested here. A separate pH 7 repeat gives 120 s (0.00833 s^{-1}); it shows variation, not a new pH value.

Explain and evaluate

pH can change how a substrate binds and how the enzyme speeds up the change. The graph does not show the active-site shape itself. It does not prove permanent denaturation. Match conditions to make a fair comparison. Samples taken 30 s apart limit how precise the time is. A shorter fixed gap narrows that window. It must be feasible and authorised.

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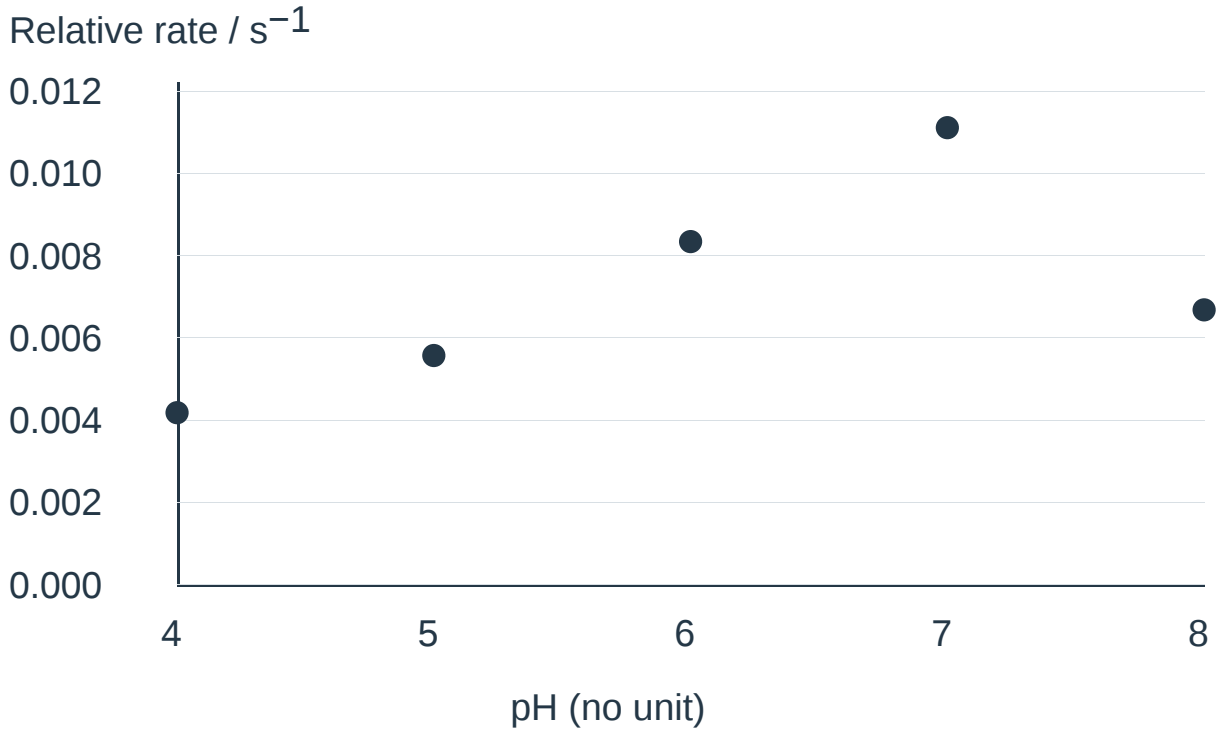
Knowledge organiser · continued

Transfer the science

Enzyme digestion changes molecules in a chemical way. It breaks large molecules into smaller ones. Diffusion is net movement of particles. They move from higher to lower concentration. That movement does not itself break starch into sugars in a chemical way.

Diagram

Dark dots show the first-series rates. The graph uses the supplied data; it is not a record of your own practical.



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Word page

Enzyme and substrate

active site — The part of an enzyme where the substrate binds. The shapes fit each other.

denaturation — A change in enzyme shape. This stops the active site binding as it should.

diffusion — Net movement of particles. They move from higher to lower concentration.

digestion — A chemical change. It breaks large molecules down into smaller molecules.

substrate — The substance that an enzyme acts on.

enzyme — A biological catalyst. It speeds up a reaction. It is not used up.

specificity — An enzyme acts on particular substrates. They fit its active site.

Measure and compare

relative rate — One divided by the endpoint time. Match the starting amounts. Use the same endpoint.

reciprocal — One divided by a number.

pH — A measure of how acid or alkaline a substance is.

endpoint — The agreed point we can see and record. We record the time when it is reached.

control variable — A factor we keep the same. This makes the comparison fair.

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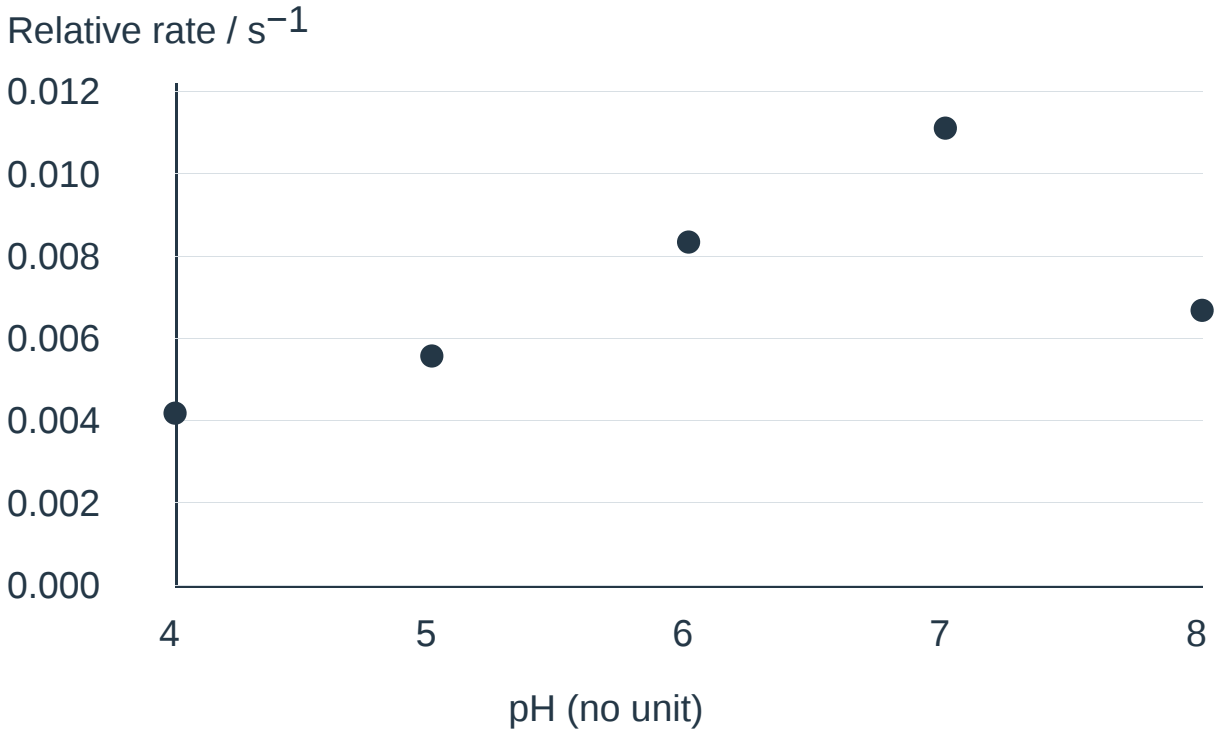
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Word page · continued

Diagram

Dark dots show the first-series rates. The graph uses the supplied data; it is not a record of your own practical.

Relative rate = 1/time. For 120 s: 1/120 = 0.00833 s^{-1}. For 90 s: 1/90 = 0.0111 s^{-1}. A shorter endpoint time gives a larger reciprocal, for matched amounts and the same stopping rule.



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We Do · response cards

Response cards

A B C

Guided example

In your book: Complete the shared task; record one decision and its reason.

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We Do · cards and recording

Use the cards

In your book: Complete the shared task; record one decision and its reason.

Choose a card, then a destination. You can change your mind.

Card 1

Reciprocal value: Endpoint 120 s gives 1/120 = 0.00833 s^{-1}.

Card 2

Direction check: The shortest time should correspond to the largest reciprocal rate.

Card 3

Highest tested rate: The largest first-series relative rate occurs at pH 7 among pH 4–8 tested here.

Card 4

Rise then fall: First-series relative rate rises from pH 4 to pH 7, then falls at pH 8.

Card 5

Subjective endpoint: Two people may judge the iodine colour in different ways. This may happen when the colour is hard to judge.

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We Do · cards and recording · continued

Card 6

Shorter interval: The approved method may let you take samples more often. Use a fixed, shorter gap. This could make the endpoint-time window smaller.

Card 7

Preserve anomaly: One repeat is far from the others. Keep it in view. Check the method. Give a reason for how you treat this result.

Destinations

CALCULATION CHECK

PATTERN / CONCLUSION

EVALUATION

In your book: Complete the shared task; record one decision and its reason.

Guided example

In your book: Record one evidence claim and explain its limit.